



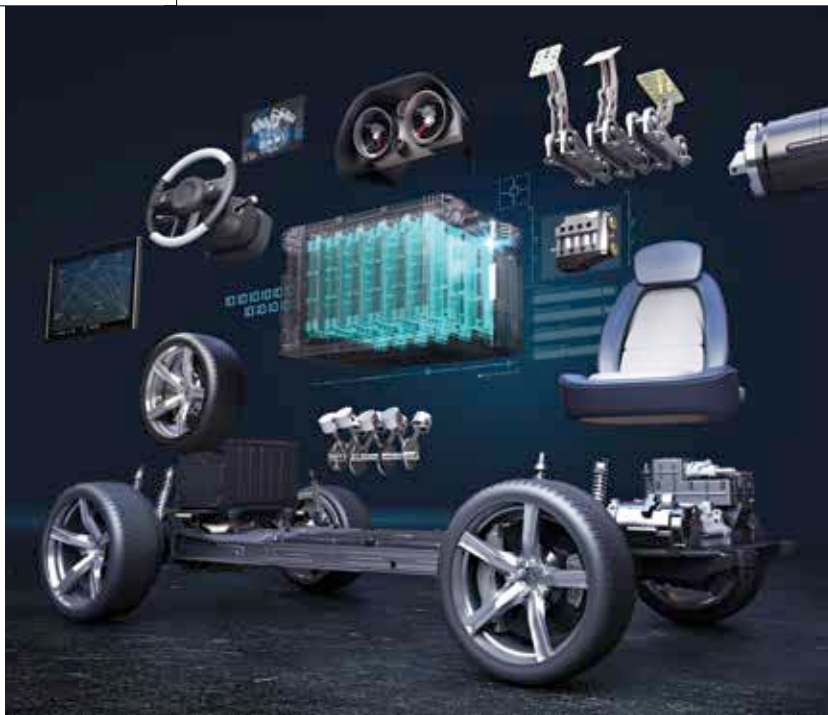
Optic nerve cell regeneration

Scientists have found a new way successfully regenerate optic nerve cells in mice. <https://digit.in/dec20-31>



T-ray technology

Radiation-based skin structure analysis could help improve treatment of skin conditions. <https://digit.in/dec20-32>



TAMING *the* LI-ION

New technologies promise to make lithium-ion batteries safer and more reliable

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Lithium-ion batteries power most smartphones right now, and are on the verge of getting even more central to everyday life, considering their potential for use in electric vehicles. Some of the problems or shortcomings of lithium-ion batteries include safety, energy density, life span and environmental impact of disposal, which is why there is so much research into superior alternatives. There are many promising options including lithium-metal,

lithium-polymer, sodium-sulfur, zinc-air, and nickel-iron batteries. Energy can also be derived from ethanol, methanol, ammonia or hydrogen fuel cells. Research into nanomaterials such as graphene and MXenes promise to improve all current and future battery technologies, including lithium-ion batteries. By incorporating nanomaterials in the batteries, they can be made longer lasting, lighter, tougher and more energy efficient. Despite all these investigations into alternatives, there is ongoing research into improving the lithium-ion batteries, the reliable option that has proven itself at scale, directed at making them safer, more

energy efficient, managing their use to optimise performance or longevity, and extending the lifespan of a manufactured product. Lithium-ion batteries are not going away any time soon, and here is the recent research into making them better.

IMPROVING SAFETY

Everyone knows that lithium ion batteries can explode and catch fire easily, including when subjected to physical stresses or exposed to flames. All lithium-ion batteries use copper and aluminium foil collectors, one for each electrode, which are thin strips by volume but make up almost 80 percent of the weight of the batteries. Researchers from Stanford and SLAC have developed collectors made up of new materials that are lighter by replacing the copper with cheaper polymer. This polymer is coated with flame retardant material, that also makes the batteries flame resistant. The fire retardant substance used in the polymer is called triphenyl phosphate or TTP. When exposed to the naked flame, the new batteries only produced weak flames that burned out within a few seconds, instead of the conventional batteries that flared up. The redesigned collectors offer several advantages for manufacturers. They use materials cheaper than the existing ones, are easy to manufacture, weigh 80 percent less, and cause a 20 percent increase in the energy density of the battery. The most important advantage is that the flame retardant material makes the batteries significantly safer to use.

MONITORING BATTERY HEALTH

Towards the end of the life cycles of smartphones, laptops and other devices powered by lithium-ion batteries, the estimated charge capacity remaining gets increasingly inaccurate the longer the device has been used. This is because all batteries, like humans, age in different ways. The diagnostic systems and tools at present do not probe the internal